

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix}$$

$$B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}$$

$$C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$B \quad C = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$
 $\begin{matrix} 2 \times 3 & 3 \times 1 \\ \text{product } 2 \times 1 \end{matrix}$
 $= 1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 0 \end{pmatrix}$

$$A \quad B = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad \begin{pmatrix} -1 \\ 0 \end{pmatrix} \perp \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 2 \end{pmatrix} & 0 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 1 \begin{pmatrix} 0 \\ 2 \end{pmatrix} & 3 \begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0 \begin{pmatrix} 0 \\ 2 \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 & -3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

Can $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ be written as a linear combination of $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$?

Does $\begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ have a solution?

Does the system of linear eqns $\begin{cases} -x_1 + 0x_2 = 1 \\ x_1 + 2x_2 = 2 \end{cases}$ have a solution?

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \quad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$DA = \begin{pmatrix} 1 & 2 \end{pmatrix} \begin{pmatrix} \boxed{-1} & \boxed{0} \\ \boxed{1} & \boxed{2} \end{pmatrix} = \begin{pmatrix} \boxed{1} & \boxed{4} \end{pmatrix}$$

$1 \times 2 \qquad \qquad 2 \times 2 \qquad \qquad 1 \times 2$

$$= 1 \begin{pmatrix} -1 & 0 \end{pmatrix} + 2 \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 0 \end{pmatrix} + \begin{pmatrix} 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 4 \end{pmatrix}.$$

Solve

$$-x_1 + 0x_2 = 1$$

$$x_1 + 2x_2 = 2$$

$$\begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$$F = \begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \quad \underline{z} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

1. Can $\begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$ be written as a linear combination of $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$?

$\hat{=}$ Can I choose numbers x_1 and x_2 so that

$$x_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix}$$

$1 \times 2 \qquad \qquad 3 \times 2$

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \stackrel{?}{=} \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$3 \times 2 \qquad 2 \times 1$

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$x_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{pmatrix} 2x_1 - x_2 \\ x_1 + x_2 \\ 0 + 2x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 \\ x_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -x_2 \\ x_2 \\ 2x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

$$\begin{cases} 2x_1 - x_2 = 3 \\ x_1 + x_2 = 0 \\ 0 + 2x_2 = -2 \end{cases}$$

$$x_1 = 1 \quad x_2 = -1.$$

$$\begin{pmatrix} 2 & -1 & 3 & 3 \\ 1 & 1 & 0 & 1 \\ 0 & 2 & -2 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1 \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}.$$

1. Is $\begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$ a linear combination of $\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$?

2. Is there a vector $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ s.t. $\begin{pmatrix} 2 & -1 \\ 1 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$?

3. Solve the system

$$2x_1 - x_2 = 3$$

$$x_1 + x_2 = 0.$$

$$0x_1 + 2x_2 = -2$$

$$\begin{pmatrix} 2 & -1 & 3 & 1 & 3 \\ 1 & 1 & 0 & 2 & -1 \\ 0 & 2 & -2 & 2 & 2 \end{pmatrix}$$

$$\begin{pmatrix} \textcircled{2} & -1 & \boxed{3} & 1 & 3 \\ 1 & 1 & 0 & 2 & -1 \\ 0 & 2 & -2 & 2 & 2 \end{pmatrix} \xrightarrow{R_2 \leftarrow R_2 - \frac{1}{2} R_1} \begin{pmatrix} \textcircled{2} & -1 & 3 & 1 & 3 \\ 0 & \textcircled{3/2} & -3/2 & 3/2 & -5/2 \\ 0 & 2 & -2 & 2 & 2 \end{pmatrix}$$

pivots.

$$(1 \ 1 \ 0 \ 2 \ -1) - \frac{1}{2} (2 \ -1 \ 3 \ 1 \ 3) \\ = (1 \ 1 \ 0 \ 2 \ -1) + (-1 \ 1/2 \ -3/2 \ -1/2 \ -3/2)$$

$$\begin{pmatrix} \textcircled{2} & -1 & 3 & 1 & 3 \\ 0 & \textcircled{3/2} & -3/2 & 3/2 & -5/2 \\ 0 & \underline{2} & -2 & 2 & 2 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - \frac{4}{3} R_2} \begin{pmatrix} \textcircled{2} & -1 & 3 & 1 & 3 \\ 0 & \textcircled{3/2} & -3/2 & 3/2 & -5/2 \\ 0 & 0 & \boxed{0} \rightarrow \boxed{0} \rightarrow \textcircled{16/3} \end{pmatrix}$$

↓ normalize leading entries.

$$\begin{pmatrix} \textcircled{1} & -1/2 & 3/2 & 1/2 & 0 \\ 0 & \textcircled{1} & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{pmatrix} \xleftarrow{\begin{matrix} R_2 \leftarrow R_2 + \frac{5}{3} R_3 \\ R_1 \leftarrow R_1 - \frac{2}{3} R_3 \end{matrix}} \begin{pmatrix} \textcircled{1} & -1/2 & 3/2 & 1/2 & 3/2 \\ 0 & \textcircled{1} & -1 & 1 & \boxed{-5/3} \\ 0 & 0 & 0 & 0 & \textcircled{1} \end{pmatrix}$$

$$\downarrow R_1 \leftarrow R_1 + \frac{1}{2} R_2$$

RREF: $\begin{pmatrix} \textcircled{1} & 0 & 1 & 1 & 0 \\ 0 & \textcircled{1} & -1 & 1 & 0 \\ 0 & 0 & 0 & \underline{0} & \textcircled{1} \end{pmatrix}$

Original: $\begin{pmatrix} \textcircled{2} & -1 & \boxed{3} & 1 & 3 \\ 1 & 1 & 0 & 2 & -1 \\ 0 & 2 & -2 & \underline{2} & \underline{2} \end{pmatrix}$

$$\begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}$$

$$\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$$

$$A = \begin{pmatrix} \textcircled{1} & -1 & 2 & -2 & 2 \\ 2 & 2 & 0 & 4 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 & -1 \end{pmatrix} \begin{array}{l} R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 - R_1 \end{array} \rightarrow \begin{pmatrix} 1 & -1 & 2 & -2 & 2 \\ 0 & \textcircled{4} & -4 & 8 & -3 \\ 0 & 1 & -1 & 2 & -1 \\ 0 & 1 & -1 & 2 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & -1 & 2 & -2 & 2 \\ 0 & 4 & -4 & 8 & -3 \\ 0 & 0 & 0 & 0 & -1/4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{array}{l} R_4 \leftarrow R_4 - R_3 \\ R_3 \leftarrow R_3 - \frac{1}{4}R_2 \\ R_4 \leftarrow R_4 - \frac{1}{4}R_2 \end{array} \rightarrow \begin{pmatrix} \textcircled{1} & -1 & 2 & -2 & 2 \\ 0 & \textcircled{4} & -4 & 8 & -3 \\ 0 & 0 & 0 & 0 & \textcircled{-1/4} \\ 0 & 0 & 0 & 0 & -1/4 \end{pmatrix}$$

$$\downarrow \begin{pmatrix} 1 & -1 & 2 & -2 & 2 \\ 0 & 1 & -1 & 2 & -3/4 \\ 0 & 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{array}{l} R_2 \leftarrow R_2 + \frac{3}{4}R_3 \\ R_1 \leftarrow R_1 - 2R_3 \end{array} \rightarrow \begin{pmatrix} 1 & -1 & 2 & -2 & 0 \\ 0 & \textcircled{1} & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\begin{array}{ll} c_3 = c_1 - c_2 & c_3 - c_1 + c_2 = 0 \\ c_4 = 2c_2 & c_4 - 2c_2 = 0 \end{array} \quad \begin{pmatrix} 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$-c_1 + c_2 + c_3 + 0c_4 + 0c_5 = \underline{0}$$

p p f f p
(free) (pivot)

$$A \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$A \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} \textcircled{1} & -1 & 2 & -2 & 2 \\ 2 & 2 & 0 & 4 & 1 \\ 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & -1 & 2 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 1 & -2 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Col relationship.

$$(A z) = 0$$

Row operation: EA.

$$(E A) z = E(A z) = E 0 = 0$$

$$col 3 = -col 1 \checkmark$$

$$col 4 = col 2 + col 3$$

$$col 4 = col 2 - col 1.$$

$$\begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 2 & 4 & -2 & 2 & 3 \\ -1 & 0 & 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} c1+c3 & c2+c3 & -c1+c2 \\ -c4 & -c4 & -c4 \end{pmatrix}$$

$\swarrow c1+c3$ $\swarrow c2+c3-c4$
 $\uparrow -c1+c2-c4$

$$= \begin{pmatrix} 0_{4 \times 3} \end{pmatrix}$$

$$\begin{pmatrix} 1 & 2 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 2 & 4 & -2 & 2 & 3 \\ -1 & 0 & 1 & 1 & -1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 3 & 5 & 1 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 3 & 6 & 10 & 2 & 2 \\ 1 & -1 & -1 & 1 & -1 \end{pmatrix}$$

4×5 4×5
 $c5$
 $c2 \leftarrow c1 + c2$
 $c3 \leftarrow c1 + 2c2$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 2 & 2 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}$$

3×3 3×4 3×4
 $r3 \leftarrow r3 + r1$

$$1r1 + 0r2 + 0r3$$

$$\begin{pmatrix} \textcircled{1} & 2 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ -1 & 0 & 2 & 2 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

$C3 = C2 - 2C1$
 $2C1 - C2 + C3$

$$S_1 \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \textcircled{1} & 2 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{pmatrix}$$

3×4 3×4

$$L_1 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ 2 & -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} \textcircled{1} & 2 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -5 & -5 & 3 \end{pmatrix}$$

$$S_2 \quad r3 \leftarrow r3 - 2r1 + 0r2$$

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -5 & -5 & 3 \end{pmatrix} = \begin{pmatrix} \textcircled{1} & 2 & 3 & -1 \\ 0 & \textcircled{-5} & -5 & 3 \\ 0 & 0 & \textcircled{1} & 0 \end{pmatrix}$$

$$\begin{cases} r2 \leftarrow r2 + 5r3 \\ r1 \leftarrow r1 - 3r3 \end{cases}$$

$$U_1 \begin{pmatrix} 1 & 0 & -3 \\ 0 & 1 & 5 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & -1 \\ 0 & -5 & -5 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -5 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$U_2 \quad r1 \leftarrow r1 + \frac{2}{5}r2$$

$$\begin{pmatrix} 1 & 2/5 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 0 & -1 \\ 0 & -5 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1/5 \\ 0 & -5 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$N \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1/5 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 1/5 \\ 0 & -5 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1/5 \\ 0 & 1 & 0 & -3/5 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$N U_2 U_1 S_2 L_1 S_1 \begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 1/5 \\ 0 & 1 & 0 & -3/5 \\ 0 & 0 & 1 & 0 \end{pmatrix} = R.$$

$$\begin{pmatrix} 0 & 0 & 1 & 0 \\ 1 & 2 & 3 & -1 \\ 2 & -1 & 1 & 1 \end{pmatrix} = S_1^{-1} L_1^{-1} S_2^{-1} U_1^{-1} U_2^{-1} N^{-1} R$$

Does $\begin{pmatrix} 0 & 0 & 1 \\ 1 & 2 & 3 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}$ have a solution?